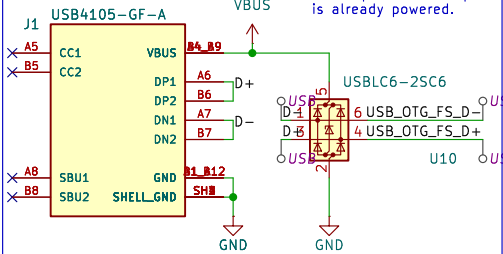
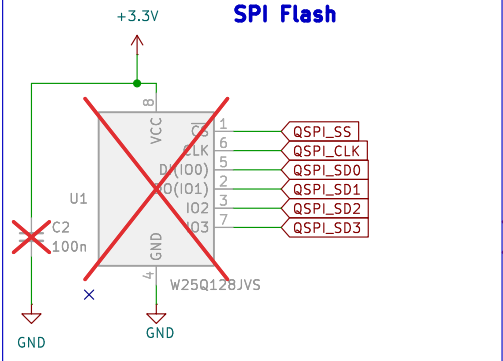


[illegible]

The diode will prevent reverse current from the board to the connected PC port when the power is already powered.

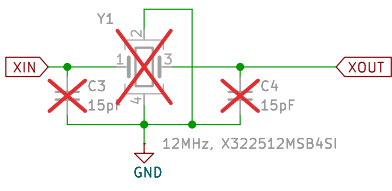


The diagram illustrates the correct and incorrect ways to connect an SPI Flash (W25Q128JVS) to a microcontroller. The microcontroller is represented by a grey rectangle with pins labeled VCC, GND, CS, and CLK. The SPI Flash is represented by a white rectangle with pins labeled 1 through 7. The correct connection is shown with a green line connecting VCC to +3.3V and a green line connecting GND to a common ground. The incorrect connection is shown with a red 'X' over the VCC and GND pins, indicating that connecting VCC to +3.3V and GND to a common ground is wrong. The diagram is labeled 'B' in the top left corner.



12 MHz Crystall Oscillator

The diagram shows a 12 MHz Crystall Oscillator circuit. It features a crystal Y1 connected between pins 1 and 3. Pin 2 is connected to ground. Pin 4 is connected to a 15pF capacitor C3, which is also connected to pin 1. Pin 3 is connected to a 15pF capacitor C4, which is also connected to pin 4. The output of the oscillator is taken from pin 3, labeled XOUT. The input is labeled XIN. The crystal is labeled 12MHz, X322512MSB4SI.

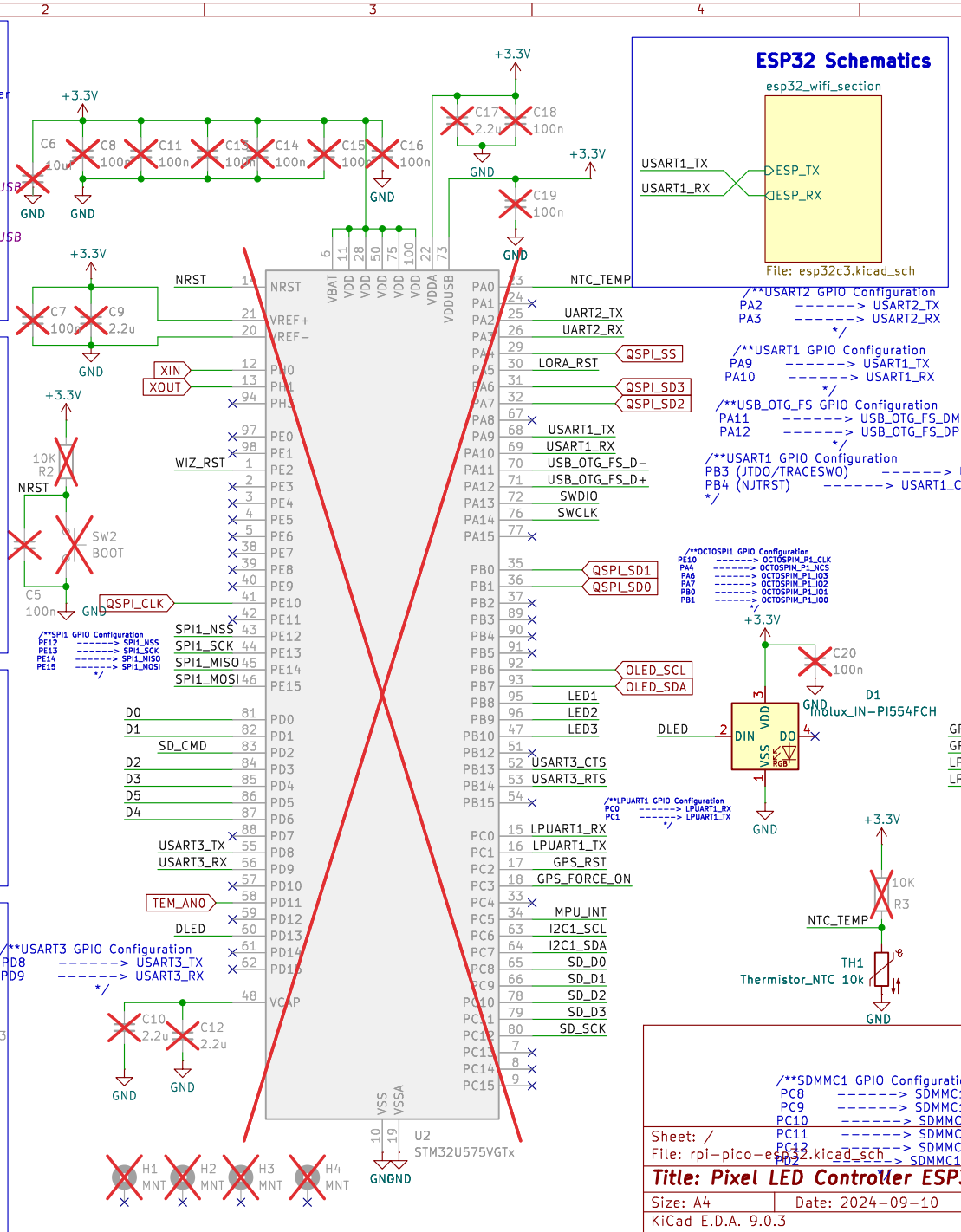
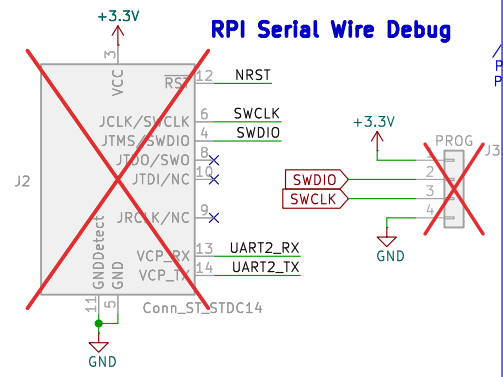


RPI Serial Wire Debug

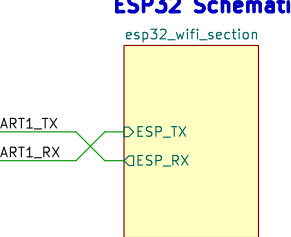
Diagram illustrating the RPI Serial Wire Debug connections. The connections are as follows:

- VCC:** +3.3V
- NRST:** (Pin 12)
- SWCLK:** (Pin 6)
- SWDIO:** (Pin 4)
- JTAG:** (Pins 8, 10, 12)
- UART2_RX:** (Pin 13)
- UART2_TX:** (Pin 14)
- GND:** (Pin 11)

The diagram also shows a connection to a **PROG** pin on a component, which is crossed out with a red 'X'.



ESP32 Schematics



The diagram shows a schematic for an ESP32 module. A yellow rectangular block represents the module, with the label "esp32_wifi_section" above it. To the left of the block, two green lines represent the UART interface. The top line is labeled "USART1_TX" and connects to a pin labeled "ESP_TX" on the module. The bottom line is labeled "USART1_RX" and connects to a pin labeled "ESP_RX" on the module. The file name "File: esp32c3.kicad_sch" is written at the bottom.

esp32_wifi_section

USART1_TX

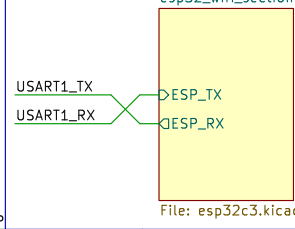
USART1_RX

ESP_TX

ESP_RX

File: esp32c3.kicad_sch

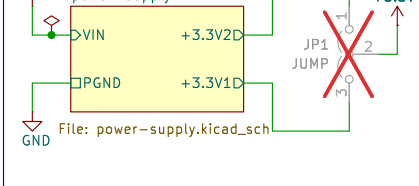
esp32 wifi section



File: esp32c3.kicad_sch

The screenshot shows a schematic diagram for a power supply. A yellow rectangular block represents the power supply component. It has three pins: VIN (labeled +3.3V2D), PGND (labeled +3.3V1D), and GND. A red 'X' is drawn over a JP1 JUMP component, indicating it should be removed. The file name 'File: power-supply.kicad_sch' is visible at the bottom.

↑ power-supply



GND

LED1 LED2 LED3

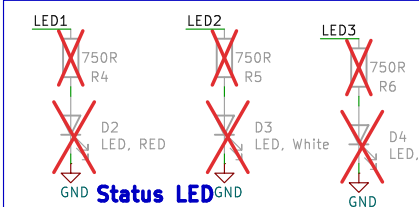
5V 5V 5V

750R R4 750R R5 750R R6

D2 LED, RED D3 LED, White D4 LED, RED

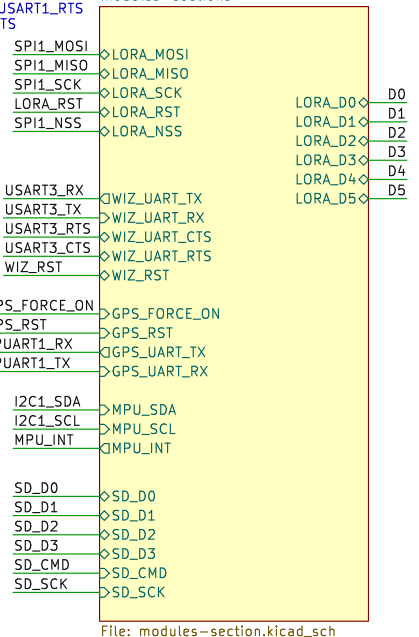
GND GND GND

Status LED



modules--sections	
USART1_RTS	
CTS	
SPI1_MOSI	↳ LORA_MOSI
SPI1_MISO	↳ LORA_MISO
SPI1_SCK	↳ LORA_SCK
LORA_RST	↳ LORA_RST
SPI1_NSS	↳ LORA_NSS
USART3_RX	
USART3_TX	↳ WIZ_UART_TX
USART3_RTS	↳ WIZ_UART_RX
USART3_CTS	↳ WIZ_UART_CTS
WIZ_RST	↳ WIZ_UART_RTS
	↳ WIZ_RST
GPS_FORCE_ON	↳ GPS_FORCE_ON
GPS_RST	↳ GPS_RST
LPUART1_RX	↳ GPS_UART_TX
LPUART1_TX	↳ GPS_UART_RX
I2C1_SDA	↳ MPU_SDA
I2C1_SCL	↳ MPU_SCL
MPU_INT	↳ MPU_INT
SD_D0	↳ SD_D0
SD_D1	↳ SD_D1
SD_D2	↳ SD_D2
SD_D3	↳ SD_D3
SD_CMD	↳ SD_CMD
SD_SCK	↳ SD_SCK

File: modules--section.kicad_sch



File: modules-section.kicad_sch

```

/**SDMMC1 GPIO Configuration
PC8 -----> SDMMC1_D0
PC9 -----> SDMMC1_D1
PC10 -----> SDMMC1_D2
PC11 -----> SDMMC1_D3
PC12 -----> SDMMC1_CLK
PC13 -----> SDMMC1_CMD

```

Sheet: / PC10 -----
File: rpi-pico-esp32.kicad_sch PC11 -----
PC12 -----
PC13 -----

Title: Pixel LED Controller ESP32+RP2040

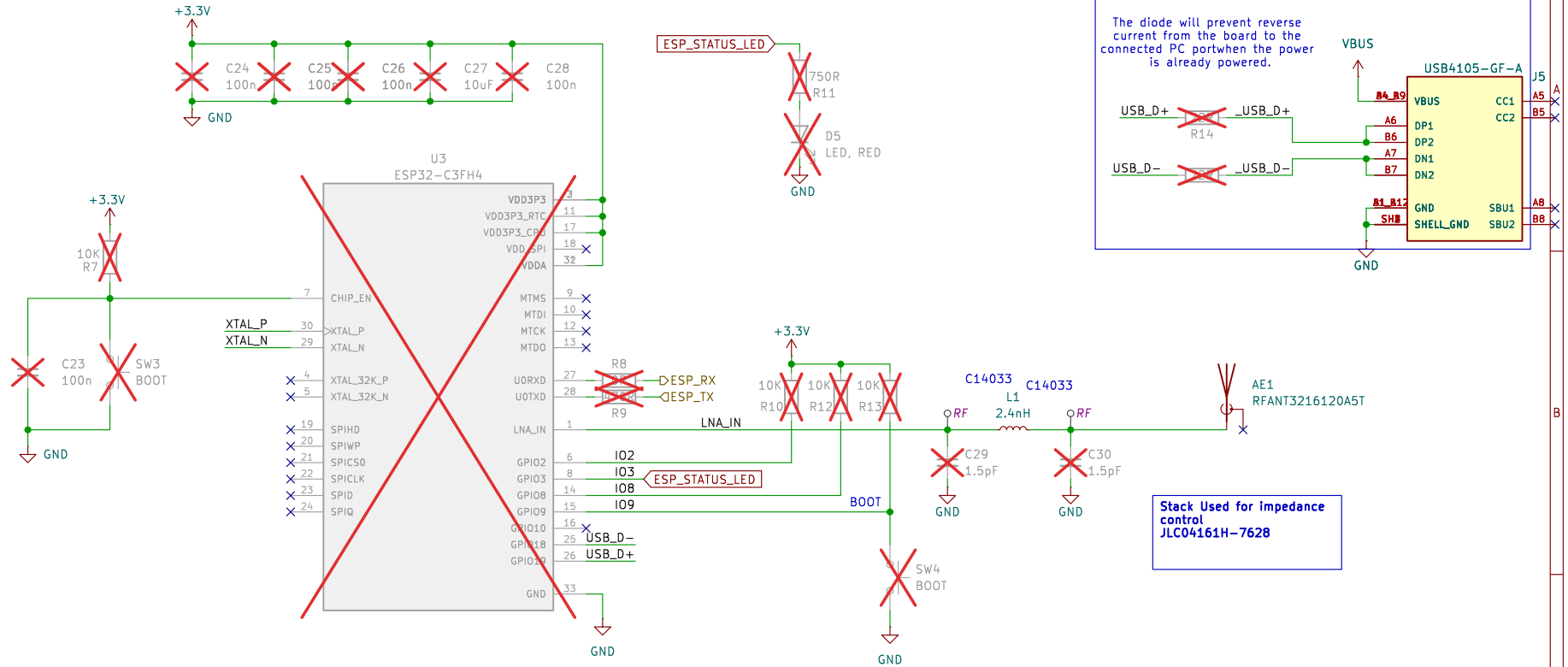
Size: A4	Date: 2024-09-10
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KiCad E.D.A. 9.0.3

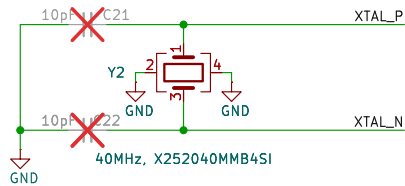
Rev: 2

Id: 1/4

ESP32 Schematics



40 MHz Crystall Oscillator



Sheet: /esp32_wifi_section/
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Title:

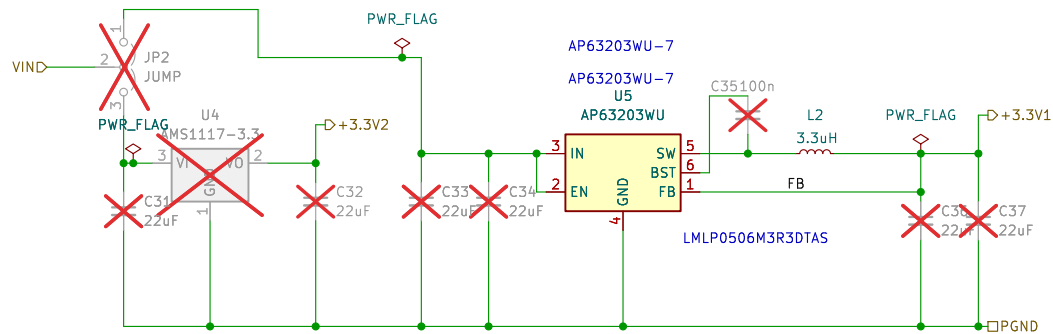
Size: A4

Date:

KiCad E.D.A. 9.0.3

Rev:

Id: 2/4



Sheet: /power-supply/
File: power-supply.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.3

Rev:

Id: 3/4

